

Advanced Haptic Directional Awareness in Virtual Reality

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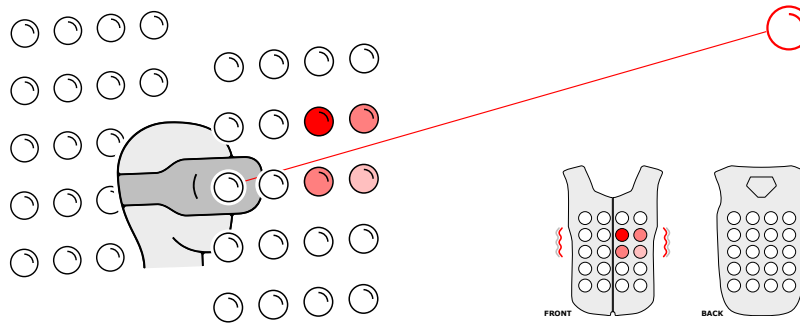
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Figure 1: Schematic illustration of direction- and angle-dependent motor activation mapping on the haptic vest.

ABSTRACT

Haptic feedback can augment human spatial awareness in virtual reality (VR) by conveying information without overloading visual or auditory channels. This paper investigates vibrotactile directional feedback as a form of sensory augmentation for navigation in immersive environments. We present a VR system using a wearable vibrotactile vest that provides head-referenced, real-time directional cues with minimal tracking requirements. A user study ($N = 42$) evaluated the system across three navigation tasks with limited or no visual guidance. Participants reliably located targets and avoided hazards using haptic feedback alone, independent of prior experience. Subjective results indicate high confidence and usability without significant cognitive interference. Eye-tracking analysis further revealed correlations between saccadic behavior, task performance, and subjective experience, suggesting opportunities for adaptive haptic feedback. These results demonstrate that vibrotactile directional feedback is an effective approach to spatial human augmentation in VR, supporting navigation and awareness when visual cues are unreliable or unavailable.

Index Terms: Virtual reality, haptics, navigation, accessibility.

1 INTRODUCTION

Haptic feedback is an essential component of the human sensory system and can convey complementary information with little cognitive interference with other modalities [9, 21]. In virtual reality (VR), which remains largely dominated by audiovisual input, haptic technology offers an effective means of augmenting spatial awareness by providing meaningful information without further cluttering visual or auditory channels [21].

In this work, we investigate vibrotactile directional feedback as a form of real-time sensory augmentation for navigation and spatial awareness in immersive environments. We present a VR system employing a wearable vibrotactile vest that delivers spatially distributed, head-referenced directional cues with minimal tracking requirements. The system supports users in a set of location-based

tasks with varying directional semantics, including target localization and hazard avoidance, under conditions where visual cues are limited or unavailable.

This approach addresses the key roles of haptics identified by Culbertson et al. [7], i.e., task or situation criticality, replacement of audiovisual cues, and enhancement of virtual interaction. Accordingly, we empirically evaluate whether vibrotactile feedback can provide clear and reliable three-dimensional directional cues with minimal encoding effort (H1), support successful task completion without significant cognitive interference (H2), and be interpreted effectively with little to no habituation across task types (H3).

2 BACKGROUND AND RELATED WORK

Haptic technologies have been widely investigated across diverse application domains, including virtual reality (VR), navigation, and sensory augmentation [1, 10, 25]. In VR, haptic feedback can enhance immersion and interaction by engaging a sensory modality beyond vision and audition, and wearable haptic devices are particularly well suited due to their seamless integration into immersive experiences. Prior work has shown that vibrotactile feedback can convey context-dependent information aligned with the virtual environment, supporting spatial and social presence.

Much existing research has focused on simulating physical contact, ranging from object manipulation using haptic gloves [5] to full-body feedback with haptic suits [12]. Such approaches have demonstrated measurable effects on realism, embodiment, and emotional response in social and navigational scenarios [6, 2, 14]. Beyond direct contact simulation, vibrotactile feedback has also been used to improve spatial awareness and navigation efficiency in cluttered or low-visibility virtual environments [15, 18], as well as for sensory substitution, for example to convey directional audio information to deaf or hard-of-hearing users in VR [17].

Haptics-based navigation has been extensively studied as an alternative or complement to visual and auditory guidance, particularly in situations involving limited visibility or high cognitive demand. In real-world contexts, vibrotactile belts and wearable displays have been shown to provide intuitive directional guidance for pilots, drivers, and pedestrians while allowing users to maintain visual attention on their surroundings [23, 22, 26]. These advantages have motivated a broad range of navigation-related applications, as summarized by Kappers et al. [13]. However, real-world haptic

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navigation is typically limited to relative two-dimensional guidance due to technical constraints.

In contrast, VR enables controlled three-dimensional navigation scenarios, such as search and wayfinding tasks, using wearable haptic devices. Prior VR-based approaches include head-mounted tactors providing vibrotactile cues [8], cheek-based stimulation mounted on headsets [20], and torso-mounted haptic vests with densely distributed tactors [24]. While some systems rely on complex tracking setups, including full upper-body tracking [18], our work demonstrates that efficient three-dimensional directional navigation can be achieved using a wearable vibrotactile vest with minimal tracking requirements.

3 MATERIALS AND METHODS

To evaluate the hypotheses introduced in Sec. 1, we conducted a controlled VR experiment ($N = 42$) in which participants completed a series of directional tasks relying primarily on haptic feedback, with little or no visual guidance. The study was approved by the Ethics Review Panel of the University of Luxembourg. Participants were recruited from the university population (43% female, 57% male; age 19–43, $M = 26.6$). Each session lasted approximately 30 minutes and took place at the VR/AR Lab's local testing area. Two additional participants were excluded due to motion sickness or previously undisclosed medical conditions.

3.1 Hard- and Software

The experiment was conducted using an HTC Vive Pro Eye headset with controllers and outside-in tracking. Haptic feedback was provided via a bHaptics X40 TactSuit vest with 40 independently addressable vibrotactile motors distributed across the front and back of the torso (Fig. 1). The VR application was developed in Unity using the bHaptics SDK for precise control of vibrotactile output. A participant wearing the full setup is shown in Fig. 2.

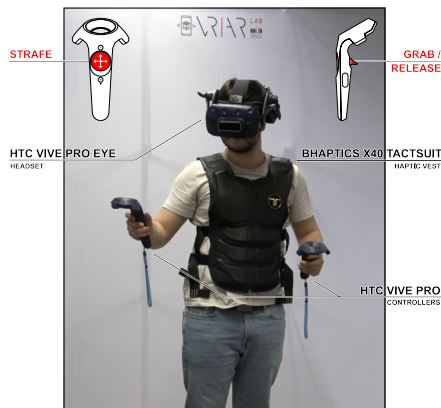


Figure 2: VR hardware and explicit controls (red).

3.2 Environment and Interaction

We designed a dedicated virtual environment consisting of multiple task zones with a deliberately minimalist visual design and minimal task-related salience (Fig. 3). After a short tutorial phase, participants completed three challenges in sequence: Floating Orbs (A), Fog Labyrinth (B), and Trapdoor Floor (C). Interaction was intentionally simple, using trigger-based object manipulation and touchpad-based locomotion, while orientation and arm movement were tracked implicitly.



Figure 3: Top-down view of the complete virtual environment and different task zones: Start/Tutorial (X), Floating Orbs (A), Fog Labyrinth (B), and Trapdoor Floor (C).

3.3 Haptic Feedback Design

Each vibrotactile motor on the vest was individually controlled to convey directional information through variations in activation location, intensity, and duration. Directional cues were computed based on the relative position of a target to the user's head orientation, using a virtual 4×5 grid of phantom reference points mapped to the vest's motor layout (Fig. 1). Motor selection and signal parameters were dynamically updated according to the user's distance and angle to the target, as well as the direction of approach.

In addition to directional cues, the system provided context-dependent environmental feedback, including rolling patterns to indicate dynamic events (e.g., gates opening) and brief vibration effects for object interaction and environmental hazards. This combination enabled both proactive navigation guidance and reactive situational awareness using a single haptic channel.

3.4 Tasks

Participants completed three location-based tasks designed to limit visual guidance and emphasize haptic navigation. In the Floating Orbs task (A), users located two visually indistinguishable targets distributed throughout a three-dimensional volume (Fig. 4). In the Fog Labyrinth task (B), users navigated a maze with severely reduced visibility to retrieve and return a target object (Fig. 5). In the Trapdoor Floor task (C), users traversed an empty room while avoiding invisible collapsing floor elements, receiving haptic warnings when approaching hazards (Fig. 6).

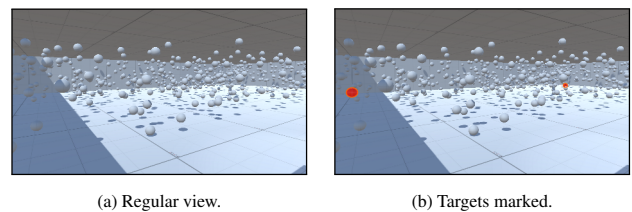


Figure 4: "Floating Orbs" zone, screenshot from entrance.

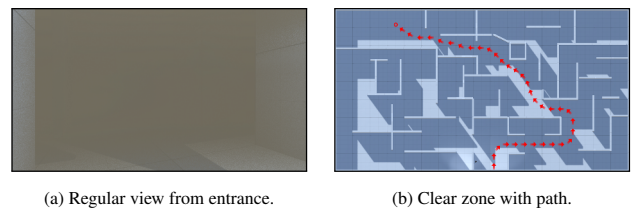


Figure 5: "Fog Labyrinth" zone, screenshot and top-down view.

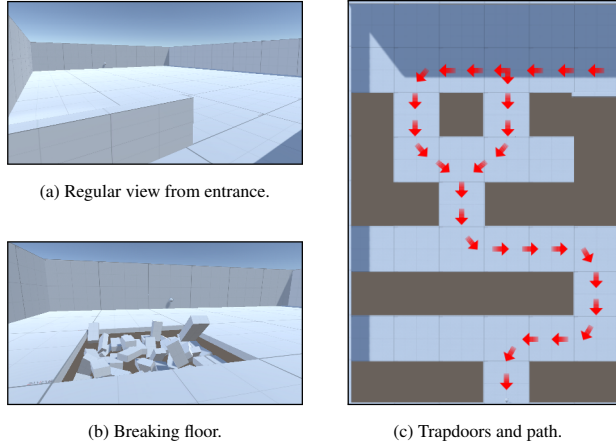


Figure 6: “Trapdoor Floor” zone, screenshots and top-down view.

3.5 Data Collection

We collected objective performance measures including task completion time, spatial trajectories, and distance to targets during haptic activation. Continuous tracking of head and body movement enabled analysis of navigation behavior, with aggregated motion traces and dwell times shown in Fig. 7. Eye-tracking data were recorded throughout to assess attentional and cognitive states.

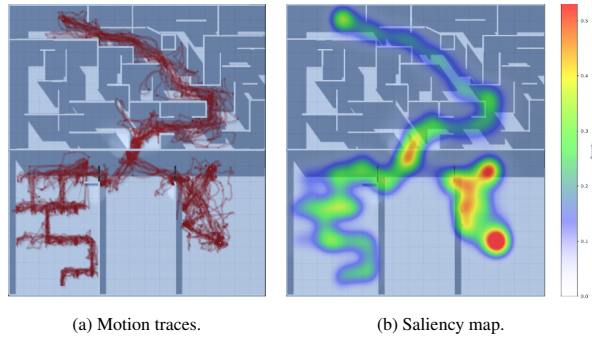


Figure 7: Aggregate user motion and dwell times during challenges.

Subjective data were collected using pre- and post-trial questionnaires assessing confidence, usability, immersion, and presence. The post-trial questionnaire comprised 15 Likert-scale items grouped by category for analysis (Tab. 1), along with optional free-text feedback. Pre-experiment data captured include demographics and prior VR, gaming, controller, and haptics experience. Additional analysis details are provided in Sec. 4 and Sec. 5.

4 RESULTS

4.1 Task Performance Measures

All participants successfully completed all challenges. The overall mean completion time was 9.6 min ($SD = 4.3$), with comparable durations across tasks: 3.3 min ($SD = 1.8$) for Floating Orbs, 3.0 min ($SD = 1.4$) for Fog Labyrinth, and 3.3 min ($SD = 1.9$) for Trapdoor Floor. When grouped by participant characteristics, completion times differed by gender and expertise. As shown in Fig. 8a, participants identifying as male completed the tasks faster on average than those identifying as female across all three challenges (Floating Orbs: 2.6 vs. 4.2 min; Fog Labyrinth: 2.3 vs. 4.0 min; Trapdoor Floor: 2.4 vs. 4.4 min).

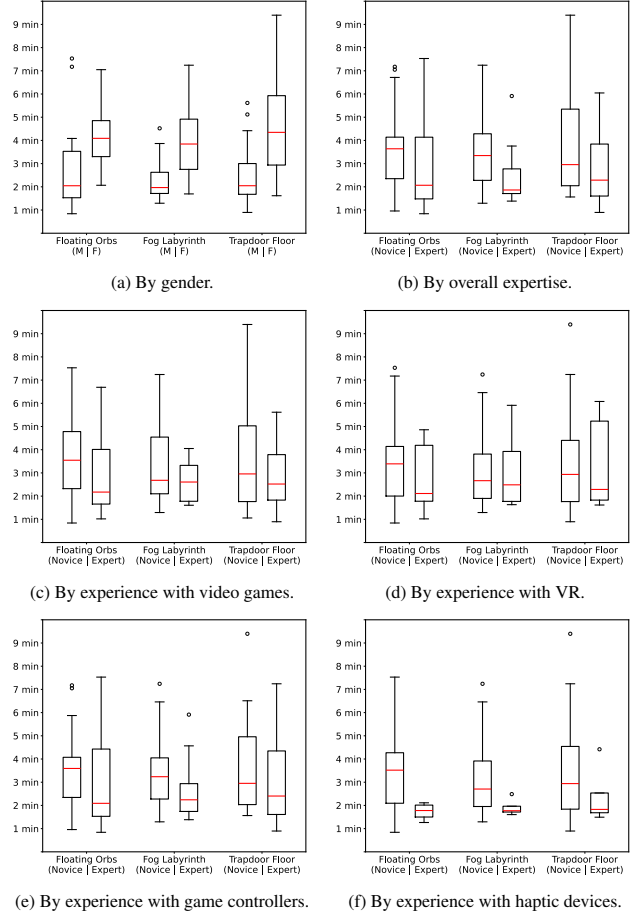


Figure 8: Participants’ average completion time grouped by gender and expertise. The red line indicates the median value.

Expertise was assessed using a composite score based on prior experience with VR, video games, controllers, and haptic devices (Sec. 3.5). The mean expertise score was 4.1 (median=4), which served as the threshold for distinguishing novices and experts. Expert participants consistently outperformed novices across all challenges (cf. Fig. 8b), with completion times of 2.4–2.8 min compared to 3.5–3.9 min for novices. Similar patterns were observed when grouping participants by specific technology experience (cf. Figs. 8c to 8f).

4.2 Subjective Measures

Post-experiment questionnaire responses (Tab. 1) are summarized in Fig. 9. Participants reported high confidence across all tasks, including the Trapdoor Floor challenge where haptic feedback primarily served as a warning signal. Most participants indicated that repeated exposure would further increase confidence. Usability ratings were consistently high: directional haptic feedback was perceived as clear, intuitive, and easy to interpret, and the controls were considered precise and intuitive. Immersion and presence were also rated highly, with minimal adjustment issues reported. While most participants did not find the haptic feedback distracting, approximately one quarter reported some degree of distraction.

Pearson correlation analysis revealed that confidence correlated with usability ($r = 0.617, p < 0.001$), presence ($r = 0.309, p = 0.046$), and performance (overall completion time; $r = -0.592, p < 0.001$). Usability also correlated with performance ($r = -0.449, p = 0.003$).

Table 1: Post-trial questionnaire.

#	Item
<i>Confidence</i>	
1	I felt confident finding the spheres in the first room.
2	I felt confident navigating through the maze in the second room.
3	I felt confident navigating the last room with the collapsing floors.
4	I would be more confident if I redid this.
<i>Usability</i>	
5	I felt the haptic feedback for directional information was clear.
6	I felt the haptic feedback for warnings was clear.
7	I felt that the directional information was intuitive.
8	I felt that the haptic feedback for navigation was precise.
9	Understanding the haptic feedback was easy.
10	I found the controls intuitive.
<i>Immersion and Presence</i>	
11	I felt immersed or involved during the experience.
12	I adjusted quickly to the virtual environment experience.
13	The haptic signals distracted me.
14	I was able to concentrate on the challenge rather than haptics.
15	I felt proficient in moving through the virtual environment.

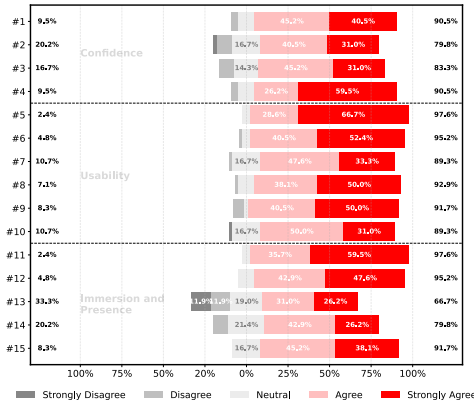


Figure 9: Participant responses (#13 inverted, negatively keyed).

4.3 Eye-Tracking Measures

Eye-tracking data were recorded throughout the experiment, including gaze direction, eye openness, and pupil size. Analysis focused on saccadic behavior due to its established links to cognitive load and attentional processes. Saccades were detected using an identification by velocity and dispersion threshold (IVDT) method [4, 11], based on gaze direction sampled at 120 Hz. Angular velocity (v_t) was computed between consecutive frames as:

$$v_t = \frac{\|\vec{n}_{t+1} - \vec{n}_t\|}{\Delta t} \quad (1)$$

Across participants, the average saccade frequency was 2.1 saccades/s, consistent with reported values for visual search tasks [16]. Saccade frequency showed significant correlations with overall completion time ($r = -0.364, p = 0.048$), confidence ($r = 0.422, p = 0.020$), and sense of presence ($r = 0.585, p = 0.001$).

5 DISCUSSION AND CONCLUSION

This study investigated vibrotactile directional feedback as a means of spatial awareness and navigation in VR under conditions of reduced visibility. The results confirm all three hypotheses (H1–H3) and provide insights into attentional behavior and individual differences relevant to wearable haptic system design.

Across all tasks, vibrotactile feedback provided clear and reliable three-dimensional directional cues with minimal tracking and encoding effort, confirming H1. Participants completed all challenges independently, even when visual cues were severely restricted, and reported high confidence (79.8–90.5%) and usability ratings (89.3–97.5%). Navigation traces indicated focused, goal-directed behavior with minimal deviation from effective paths. Consistent with prior findings, head-referenced feedback was perceived as intuitive, supporting headset orientation as a suitable reference frame for haptic navigation cues.

H2 was also supported, as haptic feedback enabled successful task completion without significant cognitive interference. Participants reported high confidence, usability, immersion, and presence, and no cognitive overload was observed. Although approximately 24% of participants reported occasional distraction, eye-tracking data suggest that increased saccadic activity reflected engagement rather than confusion. Saccade frequency correlated with performance, confidence, and presence ($r = -0.364$ to 0.585), indicating effective integration of haptic information.

The findings further support H3, showing minimal habituation requirements. In an earlier demonstrator study [3], gaze deviations from body posture had no perceptible effect on relative directional information, reducing transformation effort. Participants with limited prior experience performed comparably to experts in tasks that relied most heavily on haptic cues, particularly under low-visibility conditions. Observed performance differences were primarily associated with input device familiarity rather than difficulties interpreting vibrotactile signals, indicating broad accessibility.

The results also highlight the importance of accounting for individual differences. Performance differences across gender groups are likely influenced by vest fit, body morphology, and uniform intensity settings rather than inherent differences in spatial navigation ability [19], underscoring the need for adaptive calibration in wearable haptic systems. The vest design assumes close torso contact, which may have displaced actuators for some participants and reduced signal clarity, while uniform intensity settings may have led to under- or over-stimulation depending on body size or sensitivity. Body shape, height, and weight can all influence the clarity of vibrotactile feedback and the resolution of directional information from the motor arrays on the front and back of the vest, suggesting that future systems and follow-up studies should explore calibration procedures to account for these individual differences. One possibility, for instance, would be to adjust vibration intensity and the active surface of the haptic display based on body height (see Fig. 10). The shorter a person, the more punctual the vibrotactile output; the taller a person, the more support motors are used to enlarge the haptic feedback area.

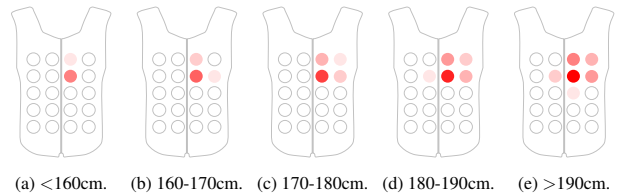


Figure 10: Haptic feedback intensities and ranges relative to height.

In conclusion, the findings demonstrate that vibrotactile directional feedback is an effective form of spatial human augmentation in immersive environments, supporting navigation and awareness with low cognitive load and minimal training. Beyond VR, the approach shows promise for mixed-reality applications such as training, operational support, and assistive technologies. Future work should explore adaptive haptic systems that integrate physiological signals and calibration to better account for individual differences.

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